



Nozzle Geometry Optimization for Reducing Jet Noise in Aircraft Engines

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Summary

The airline industry continues to grow, ICAO estimated that it will increase by about 4.3% every year, but the aircraft noise hindering the further development of air transport including, public concerns about the noise damage have intensified, government policy and airport regulation have been strict, and airline operation cost like noise tax have been increased. Jet noise is the main noise source of aircraft, especially take-off noise, which is the focus of this paper. To begin with, the mesh quality, turbulence model and boundary condition setting is explored and comparing with experiment result for the CFD validation purpose. To reduce the jet noise, two reduction methods, fan flow deflection method and chevrons method are used. For the fan flow deflection approaching, the 40° and 60° wedge-shaped deflectors are inserted into the nozzle individually. The simulation results shows that the wedge can tilt the bypass jet downwards and the large angle wedge can reduce about 5dB noise with little thrust lost. Two features, chevrons length and penetrated angle, are explored. The length of chevrons is related to noise reduce and 40mm length chevrons used in this paper can reduce 3 dB noise. for penetrate angle, the 20° angle of chevrons can slight reduce the noise, but the -20° angle can't reduce noise. However, small penetrate angles should be simulated in the future, and hot jet instead of cold jet should be used in the future simulation.

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NOMENCLATURE

Symbol	Quantity	Unit
x	streamwise coordinate	m
y	transverse coordinate	m
U	mean value of velocity	m/s
Lw	lift of wedge	N
q	dynamic pressure	Pa
Cl	lift coefficient of half angle of wedge	NA
S	area of wedge which flow pass	m ²
Uj	jet velocity at jet exit, assuming fully expanded flow	m/s
Dj	diameter of single-jet nozzle, where jet exit	m
P	acoustic power	w
Ds	diameter of bypass nozzle	m
k	turbulence energy	w
Dp	diameter of core nozzle	m

ACRONYMS

Acronym	Meaning
DES	Detached eddy simulation turbulence model
SST	k-omega (k- ω) SST turbulence model
SPL	Sound pressure level
OASPL	Overall sound pressure level
FW-H	Ffowcs-Williams equation

1. INTRODUCTION.

1.1. Background

The air transport industry is developed quickly and play an important role in modern life and world economy. in 2017, the airlines in the world carried about 4.1 billion passengers and 56 million tons of cargo. This provided 65 million job opportunities and occupied the 3.5 percent of the GDP worldwide. In addition, the demand for air transport will grow continually, International Civil Aviation Organization [1] estimated that it will increase by about 4.3% every year over the next 20 years, the figure 1 shows the detail aviation development trend.

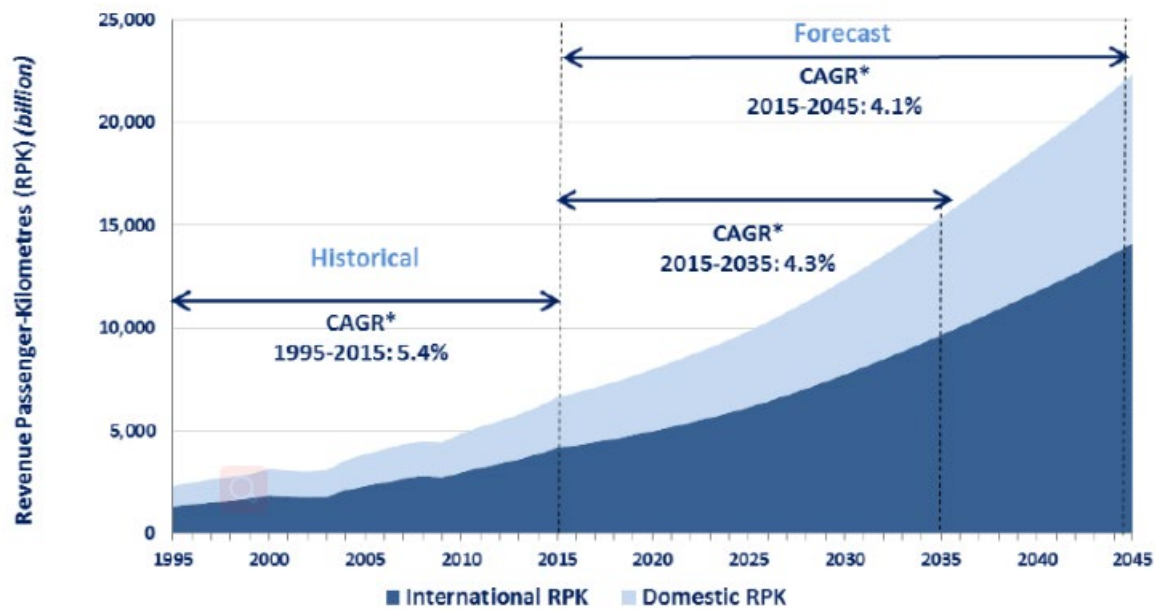


Figure 1. the developing trend of air transport [1]

Although the unprecedented COVID-19 has made a huge recession in air transport, cargo maintains high growth and global aviation passenger shows a recovery since 2021 [2]. However, with the development of air transportation, aircraft noise has become a major factor hindering the further development of air transport, and public concerns about the corresponding health and environmental impacts have intensified. A survey of residents around the airport found that long-term exposure to noise has a huge impact on human health. For instance, the aircraft's noise can cause community annoyance, disrupt sleep, children's cognition, and increase the possibility of cardiovascular disease [3]. Due to its environmental impact and public concerns, A growing number of governments and local governments are proposing noise compensation policy to local communities. This may include assistance with relocation costs and provide acoustic insulation to noise-sensitive buildings. For example, Heathrow airport launched a scheme to provide noise insulation to homes within the 55dB noise contour and offered full coverage of insulation costs to properties closest to the airport with higher noise levels. The program was estimated to cost approximately £700 million [4]. In addition, governments are going to introduce a noise charge to push the air transport industry

develop towards environment friendly way. These large amount of compensation and tax increases the overall cost of airlines, to reduce environment cost, airlines must find approaches to manage aircraft noise and aviation industries should design more environmentally friendly aircraft to meet the future trend. The aviation industry has set an aspirational target to reduce perceived aircraft noise emissions by 65% compared to typical new aircraft in 2000, by the year 2050 [5]. The current initiatives on aircraft noise are focus on 4 aspects: reduction of noise at source, noise abatement operational procedures, operation restrictions and airport land-use planning. Reducing noise from the source is the key to solving noise problems, all other methods are sacrifice airline profits in exchange for noise reduction and have many limitations. For example, according to some regulations on the operation restrictions, aircrafts that does not meet noise standards cannot take-off and landing in the airport, reducing the number of take-offs and landings to reduce noise, and adjusting the take-off and landing time [6]. They limit the ability to plan and schedule flights in an optimal manner and also hinder passenger flight connections and it could also cause avoidable flight congestion during night and early morning hours. In addition, it can result in airports consistently being less than optimal throughput with a corresponding increase in operating costs. Therefore, reducing noise from the sound source is the most essential method. It not only meets the aviation industry's pursuit of high-advance technology, but also reduces the cost of airlines in dealing with noise and meets the public's expectations for green aviation.

There are various sources of noise associated with aircraft, including the airframe, landing gear, landing flaps, and engine, it is typically the engine that produces the highest levels of noise. Jet noise is a particularly significant type of noise generated by the aircraft engine, particularly during take-off [7]. A new and effective method for exploring jet noise has been provided with the remarkable progress in the field of Computational Fluid Dynamics (CFD). By utilizing CFD technology, researchers can simulate the flow, turbulence, and sound waves without the limitation of the lab condition, providing an in-depth exploration of the underlying mechanism of jet noise.

1.2. Aim and Objectives

The aim of this project is to investigate jet noise by utilizing Ansys fluent to analyse the effects of modifying nozzle geometry. The noise control method fan flow deflection method and the chevrons method will be explored.

The objectives of the project were as follows:

- Validate the method: Simulation the single-steam jet exit from the nozzle with different turbulence model like SST, DES in 2D and 3D dimension. Obtaining the velocity, turbulence intensity and potential core length simulation result with different conditions such as different jet velocities and temperatures. Compare the result with experiment data on the literature. Choose the proper turbulence models and CFD setting up and control the error within 20%.

- Using the CFD to analyse how fan flow deflection method effects on the jet noise. Check whether inside wedge-shape deflector can tilt the upside by-pass flow downwards to generate low-velocity region to reduce noise, in addition, 20° and 40° angle wedge deflectors will be applied to compare the jet noise reduce effect.
- Exploring the effect of chevrons shape on noising reducing with the help of the CFD on the co-axial nozzle. Simulating the models with different length, penetration angle of chevrons to obtain the results like velocity contours and sound pressure level. Analysis of the influence of different types of chevron shape nozzles on jet noise.

2. LITERATURE REVIEW

2.1. Background and source of jet noise

In the early 1950s, Lighthill proposed the acoustic analogy theory and was seen as the beginning of the jet noise research. The noise is generated by the small eddies, these small eddies are known as fine-scale turbulence structure recently. The basic of this theory is to express the compressible equations of motion as a form of acoustic wave propagation and the relationship between the acoustic power and jet velocity is $P \sim U_j^8$. Later many scientists attempted to improve the theory. For example, Ffowcs Williams extended the argument with source convection theory which for the high-speed jet. The relationship become $P \sim U_j^3$. In general, this theory assumed the sound source as quadrupole sound source and the prediction largely depend on the turbulence information [8]. In the 1970s, the Crow and Champagne (1971), Brown and Roshko (1974), and Winant and Broward (1974) found that the large-scale eddies in shear flows are also a source of jet noise. Therefore, the jet noise is made up by large-scale turbulence and fine-scale turbulence and they are the product of interaction between jet in the shear layer and the ambient air. For other noise source, screech tones and broadband shock noise are not interested in the project. The large-scale eddy is intense and directional and affects downstream direction. The fine scale one is mostly uniform and affects the lateral and upstream directions. For large turbulence, it is more deterministic than fine turbulent motion and domain the dynamic and mixing processes [9]. The investigation of the spatial development of small- scale and large-scale eddies indicates that small structure eddies have a shorter streamwise lifespan and are found close to the nozzle, while large scale eddies persist longer and occur over wider regions of the shear layer or jet. Through the spectral behaviour of the sound field, small-scale eddies generate higher-frequency sound and lower-frequency sound is caused by large-scale [10].

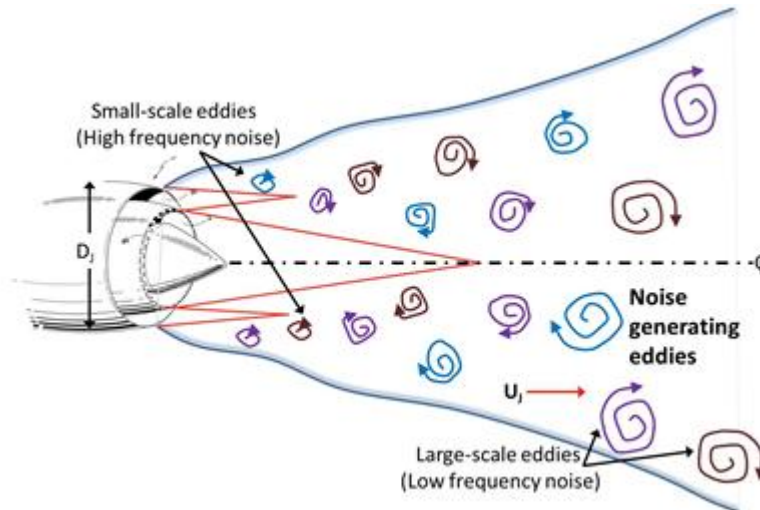


Figure 2. the small-scale and large-scale turbulence structures [11]

2.2. Jet noise control technology

As mentioned above, the jet noise is caused by the high-speed jet exits from the nozzle and the low-speed freestream in the atmosphere and from the equation $P \sim U_j^8$, the simplest way to reduce the noise is to reduce the jet exhaust velocity, but according to relationship between the thrust and velocity $T \sim U_j^2$, the balance should be made. Considered the high frequency noise quick dissipation characteristic and the human audible frequency range is from 20Hz to 20kHz, for some high frequency noise beyond the region, the human can't hear. Hence, Ahmad [12] proposed a strategy for noise control is to prioritize the reduction of low-frequency noise, but the high-frequency noise often tends to increase as a result.

2.2.1. Fan Flow Deflection Method

For turbofan engine, by changing the shape of nozzle trailing edge and nozzle type can effectively change the length of potential core and the size of turbulence to suppress jet noise. The core nozzle flow was the main noise source and increase potential core length in the bypass nozzle can reduce convective Mach number which can govern the Mach wave emission and reduce the noise [13]. Papamoschou [14] did the experiment in fan flow deflection method. The deflectors were used to tilt the bypass flow and mean flow to cause the misalignment of the two flows that create a thick, low-speed secondary core. The result showed, in shear layer, the convective Mach number of turbulent can be decreased. The shape of deflectors are vane types and wedge types, and the tilt effect of the deflectors are defined as a total deflection angle of the bypass jet from all of the deflectors. The most important fact to calculate the turning effect is force generate by deflectors, for the wedge-shape deflectors, the force can be calculated in $Lw = Cl(\alpha w)qS$, where the lift coefficient is related to the half angel of the wedge, q is the dynamic pressure on the nozzle outlet, and the S is dependent on the material solidity and wedge geometry. The use of a wedge-shaped deflectors to reduce noise in high-bypass engines cause a static-thrust loss estimated at around 0.5%. To mitigate this loss, the wedge can be designed as an actuated device that is deployed only when noise reduction is necessary. Such as when aircraft is take-off. and

stowed during cruise. The aerodynamic penalties of the wedge are limited as the pressure change induced by the wedge-shaped deflectors is small and is not expected to differ significantly from the ambient pressure [15].

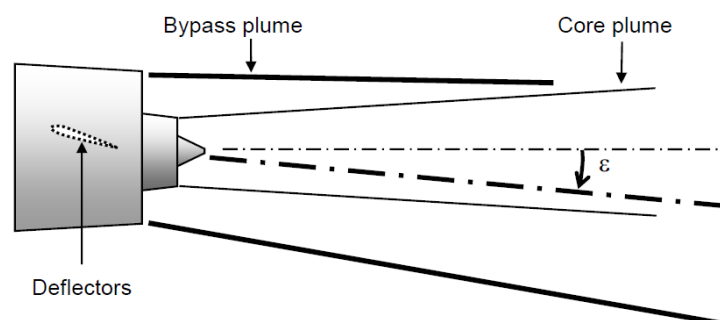


Figure 3. Concept of fan flow deflectors methods [15]

2.2.2. Chevrons Method

Ahuja [16] explored the boundary layer at the nozzle outlet and found that the fully developed boundary layer can cause higher jet noise, and changing the nozzle shape can limit the boundary layer to develop and change the size of jet radiation noise. Chevron is one of the nozzle geometry scientists mainly focus on and detailed mechanisms were still lack of understanding [7].

The general effect of the chevrons is it will enhance the mixing of the shear layer by generate the streamwise vortices, the increased mixing results in changes in the noise level over specific frequency ranges. typically, chevrons can reduce low-frequency noise at downstream locations with low polar angles, while simultaneously increasing the high-frequency noise levels perceived by an observer orthogonal to the jet axis (polar angle 90). The object of chevrons design is to decrease low-frequency noise as much as possible while minimizing any significant increase in high-frequency noise. However, the impact of thrust should be minimized while reducing the noise, experimental data shows that a specific chevron nozzle design can achieve noise reduction during take-off while producing less than 0.5% thrust loss during cruise [17]. Bridges and Brown [18] investigated the various parameter on a series of chevron nozzles about the flow and noise characteristics. These parameters include the number of chevrons, chevron penetration. and chevron length. By changing geometry of chevrons. the spacing, strength and distribution of axial vorticity will be changed. Callender et al. [19] studied the number of chevrons in a nozzle and found this has minimal effect on the noise level of the jet. This can be attributed to the close proximity of adjacent chevrons, which results in the cancellation of streamwise vortices generated by one chevron with those of opposite sign generated by the next chevron.

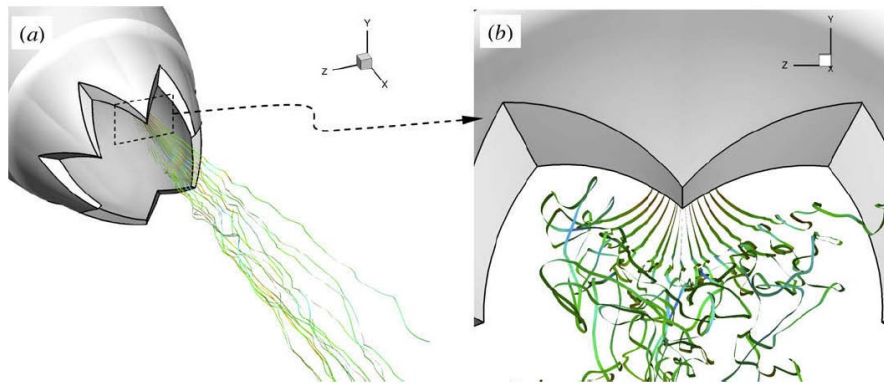


Figure 4. 'Instantaneous streamlines' near a chevron tip [17]

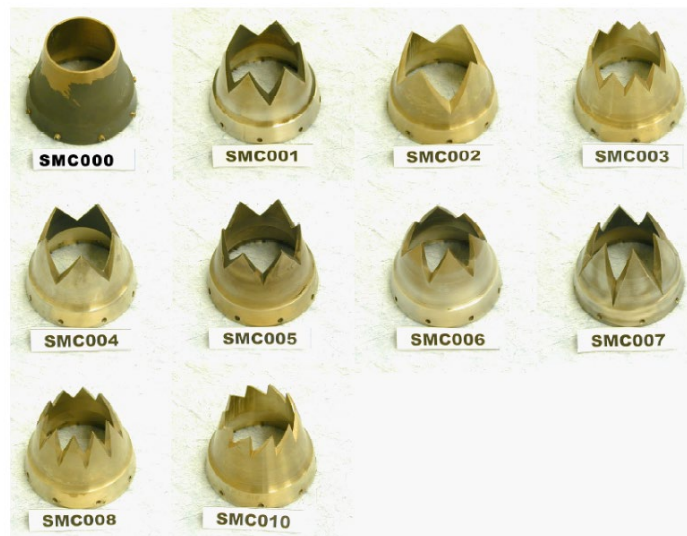


Figure 5. different types of chevron nozzles [18].

2.3. FW-H equation and Turbulence models

The Ffowcs Williams-Hawkings (FW-H) equation is a mathematical model used to predict the acoustic radiation of unsteady flows. Since this equation is based on the conservation laws of fluid mechanics rather than the wave equation, it has advantages in the analysis of aeroacoustics. The FW-H equation is an exact reordering of the Navier-Stokes equation and the continuity equation, presenting a non-homogeneous wave equation with two area source terms, a thickness source and a load source, and a volume source term, but the volume source is negligible in many subsonic applications [20]. The FW-H equation has been widely accepted by URANS, DES and LES flow solvers and the equation has been used with great success for jet noise prediction. Brès and Lele [21] have used Hybrid RANS-LES with FW-H equation to predict jet noise, in their research, the noise source set up is explored. A permeable FW-H surface is used to surround the noise source. Noise from the volume source is usually ignored because of the huge computational resources required for volume integration. The location of the FW-H surface is critical because if the FW-H surface is inside the turbulent flow, the data

collected on its surface will not include the sound produced by the external turbulence and spurious sound will generate when turbulent flow pass. However, it is unavoidable that the turbulence through the downstream outflow surface, Shur et al. [22] proposed three different downstream end treatments, open surface, close surface and averaging surfaces S5, in the figure below, for the open surface, the only surface exited is S1 and adding an outflow disk is close surface. The surfaces used for average is S5, several FW–H surfaces located at different axial locations with same intervals.

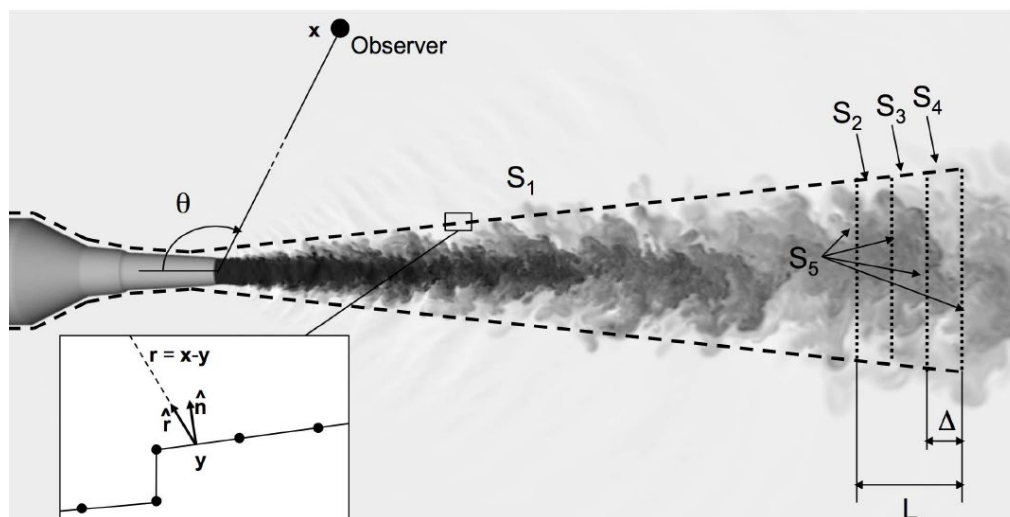


Figure 6. Example of FWH surfaces [23].

In the case of a defined noise source, the acoustic result depends on the distance from the receiver to the nozzle outlet, Turbulence Research centre in Stanford university [23].has proved through experiments that it is appropriate to set a distance of 100 times the nozzle diameter.

The jet noise prediction method based on RANS can accurately predict the influence of different design nozzle flow changes on the far sound. Based on the $k-\epsilon$ turbulence model, the prediction method is based on the assumption of isotropy, so it can accurately predict the sideline and upstream noise dominated by small-scale turbulent noise but cannot accurately predict the downstream noise of jet flow dominated by large-scale turbulent structures [24]. The Detached Eddy Simulation (DES) turbulence model represents a hybrid Reynolds-Averaged Navier-Stokes – Large Eddy Simulation (RANS-LES) approach to the simulation of complex, separated flows. This model addresses challenges associated with RANS and LES models, for example, high computational cost of LES in boundary layers and limited capability of RANS models to solve unsteady flow structures and determine the turbulence length scale. By employing a functional simulation type switching strategy in DES model, the DES model operates in both RANS and LES modes. The RANS mode is primarily used for flow field domains that can be treated as relatively steady, such as in boundary layers. The LES mode, on the other hand, is applied to the dominantly unsteady region [25]. The DES model provides a more accurate and less computational cost method to turbulent flows and acoustic waves in separated flows. Xu and Li [24] found that the far-field assumption of infinity has a greater

impact on the noise downstream of the jet, and the closer to the quiet zone of the jet, the greater the deviation of the prediction results caused by the far-field assumption. Therefore, for the downstream noise prediction of jets dominated by large-scale turbulent structure noise, local parallel flow assumptions and far-field infinity assumptions should be avoided. Tide and Babu [26] use of SST $k-\omega$ model with the FW-H method to predicting jet noise with and without chevrons, the SPL result shows that it can predict the downstream noise well but has some issue for upstream noise. With the help of FW-H equation, Sinha and Erwin [27] use Hybrid RANS-LES method to predict jet noise with different nozzle types and the result is agree with the experiment.

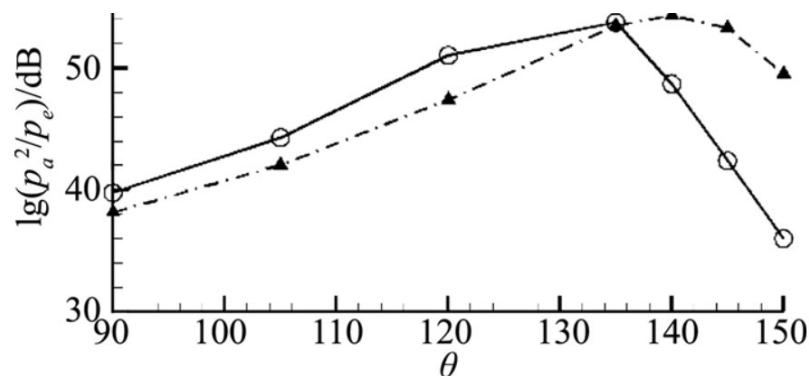


Figure 7. Effect of far-field assumption on prediction of jet noise [24]

3. METHODOLOGY

As a CFD project. The project began with validating CFD model result with the experiment data [28] published on the NASA website. Then building a turbofan CAD model as a basic model, the aim of the project is found the general trend, to reduce the difficult relatively simple will be applied. On the basis of the basic model, using the fan flow deflection and chevrons approaches individually to reduce jet noise.

3.1. CFD Validation

The validate process divided into four parts. For the first part, the mesh quality was explored, several 2D mesh from coarse to fine was generate, to find optimal mesh set up strategies in the result and computation speed. For the second part, the boundary condition set up was checked, especially for the freestream set up. Ensure that freestream would not reverse near the inlet boundary. For the third part. 2D steady SST turbulence model with different Mach number was running, turbulence intensity and velocity contours are compared with experiment data and some plots like x/D_j vs U/U_j along the centreline, y/D_j vs U/U_j at $x/D_j=2,5,10$ and x/D_j vs k/U_j^2 was generated to help comparing. For the fourth part, the transient 3D unsteady SST and DES turbulence model were running and compared with experiment result same as the third part.

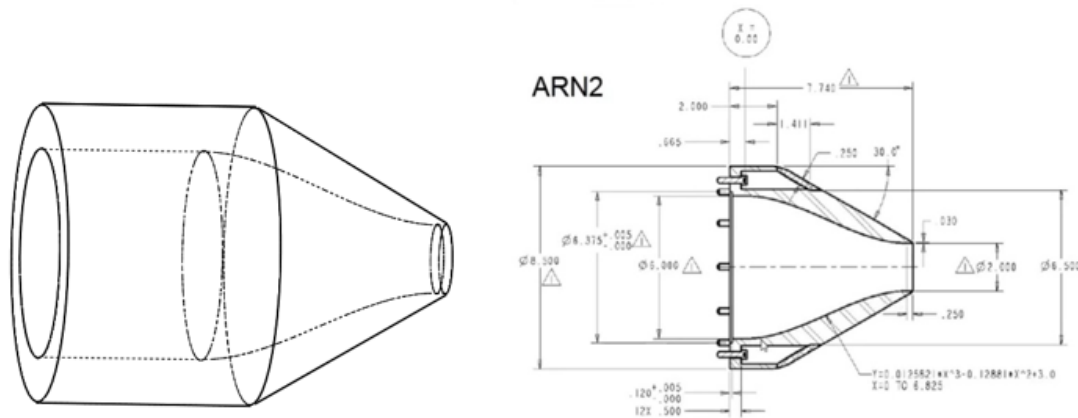


Figure 8. CFD validation model and ARN2 nozzle geometry [28]

The type of nozzle used in the validation is ARN2, Because this type of nozzle is widely used in experiments, and experimental data related to the validation model can be found. For the ARN2 model used in the validation, some modifications are made, the inlet pipe is extended and outer wall shape is simplified. However, the characteristics of the inner wall shape of the nozzle were kept unchanged to ensure that the flow behaviour inside the wall remained consistent with the experiment. The function of inner nozzle wall shape is $y=0.0125821x^3-0.12881x^2+3$.

For 2D case, element sizing with number of divisions is used to generate the mesh. The strategy for the mesh setting up is using fine mesh near the nozzle area, especially the area close to the nozzle wall and behind the nozzle (where jet exit and mixed with freestream). These areas have complex geometry such as the nozzle wall, and the region where high-speed jets pass through, and mixed ambient flow needs high accuracy and any small changes in the region may cause different results. Therefore, small mesh is used. For the other regions, like the top regions and the region far away from the nozzle, only freestream exist, or the region will have little effect on the result, as these regions are not the main focus on our research, so relatively coarse mesh is used to improve solution speed. For the detail size of mesh, this will explore in the validation process. In 3D case, for the meshes in the nozzle, small size unstructured tetrahedral mesh is used as nozzle shape is complex and for the other zone, the structured hexahedral/ prismatic mesh is used, the overall mesh generate method for the model is multizone method. The strategy of 3D case is as same as 2D case, fine mesh is used at nozzle and near nozzle area, and coarse mesh is used in the region far away from the nozzle. In addition, the periodic boundary will be used in the next steps, The premise of using periodic boundary is the meshes in the high and low sides should be symmetric, so the match control is used.

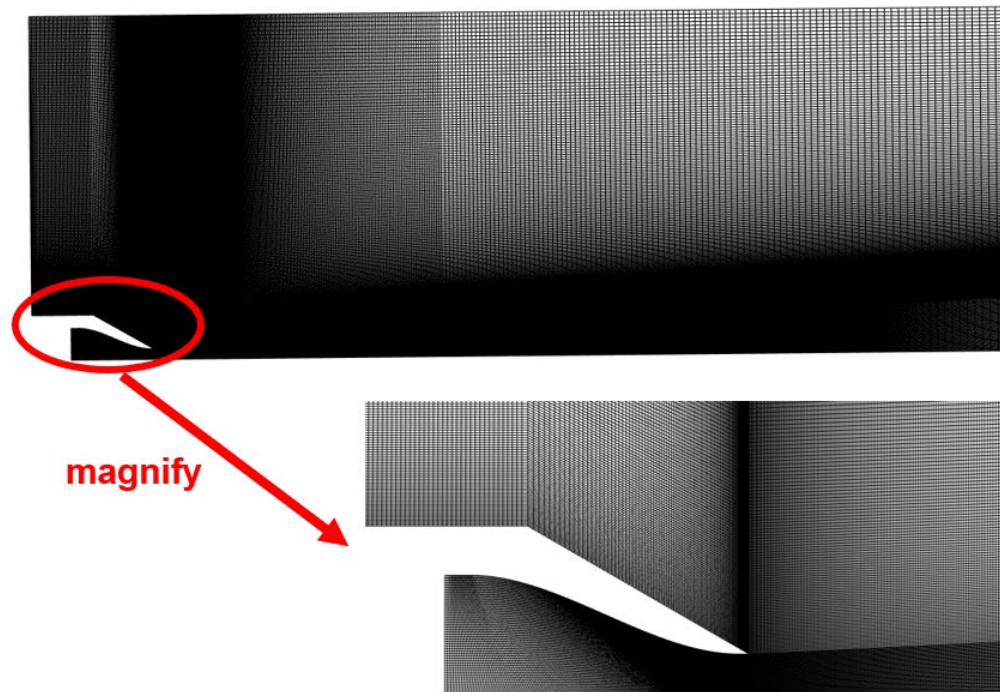


Figure 9. 2D mesh used in the CFD

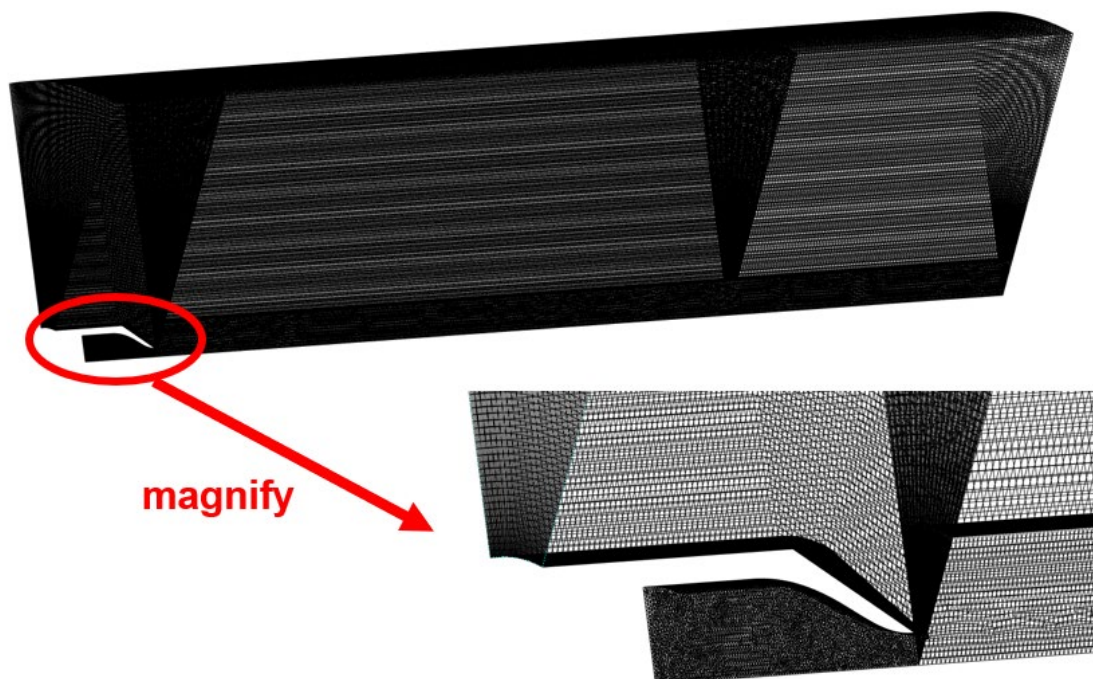


Figure 10. 3D mesh used in the CFD

For the CFD simulation set up. In the 2D case, to save the simulation time, the axisymmetric method is used to run half of the model and the solver is steady and pressure based. Considering only cold jet will run on the simulation, the ideal gas with Sutherland viscosity is chosen. For the boundary conditions, the freestream is set as velocity inlet type with Mach number 0.01. The velocity in the jet also set as velocity inlet type with $Ma=0.51$, 0.99 respectively. The pressure outlet is chosen, and the pressure is as same as ambient pressure. In the 3D case, consider the nozzle model is rotationally symmetric model and can be divided into several same parts, the periodic method which can run one divided part and the simulation result is as same as simulation the whole model is chosen. Here, the model is divided into 12 parts, with each part forming a 30-degree angle around the selected axis. This method reduces the number of elements to $1/12$, which greatly improves the speed of simulation. The boundary set up, expect the periodic boundary, is similar with the boundary set up in the 2D case. To run the 3D transient case, run steady case first, then change it to transient to save time and the time step is set by minimum mesh size / jet exit velocity first and adjusting depend on simulation condition.

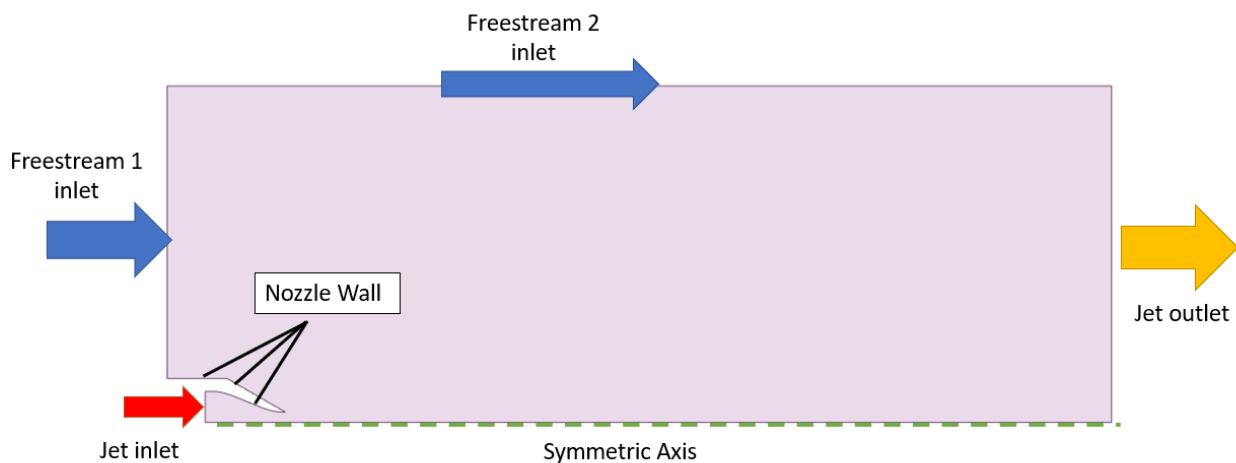


Figure 11. 2D case boundary condition set up

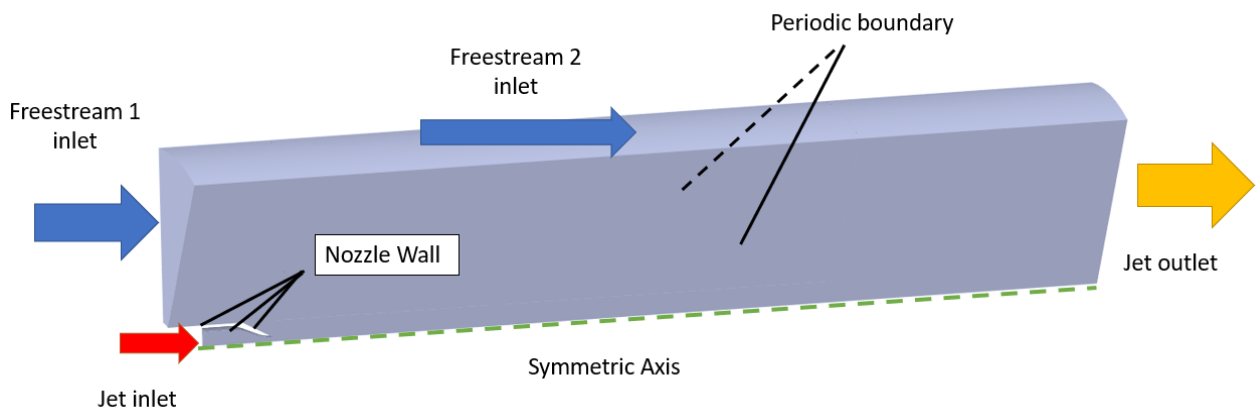


Figure 12. 3D case boundary condition set up

Table 1. Boundary condition set in validation model

Name of Boundary Condition	Type	Value
Freestream 1 inlet	Velocity inlet	0.01 Ma
Freestream 2 inlet	Velocity inlet	0.01 Ma
Jet inlet	Velocity inlet	0.51 / 0.99 Ma
Jet outlet	Pressure outlet	101325 Pa
Nozzle Wall	Wall	NA
Axis	Axis	NA
Periodic boundary (Only 3D)	Periodic	NA

3.2. Basic model used in the program

Considering that the nozzle selected as a control group for comparison with other geometrically optimized nozzles should have the characteristic of easy modification. And the research purpose is to find a correlation between the jet noise and nozzle geometry, Thus, a simple basic shape is sufficient for this investigation. Therefore, the coaxial nozzle in the figure above is selected. Some detail information about nozzle shows below. The exit diameters of the nozzle are $D_p=55\text{mm}$ for the primary and $D_s=100\text{mm}$ for the secondary and the area ratio of the nozzle is 2.28. The wall thickness of both core and bypass nozzles are 10mm. the total length of the nozzle is 158mm, at the start 108mm, the nozzle walls of the core and the bypass are parallel and the second half of 58mm is the nozzle lip with taper section.

Table 2. Nozzle size feature

Name of the geometry	Value
primary nozzle diameter	55mm
secondary nozzle diameter	100mm
wall thickness	10mm
Parallel section length	108mm
tapered section length	50mm
area ratio	2.28

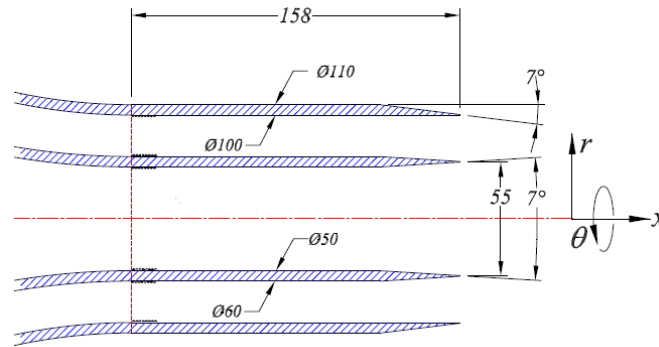


Figure 13. 2D co-axial model

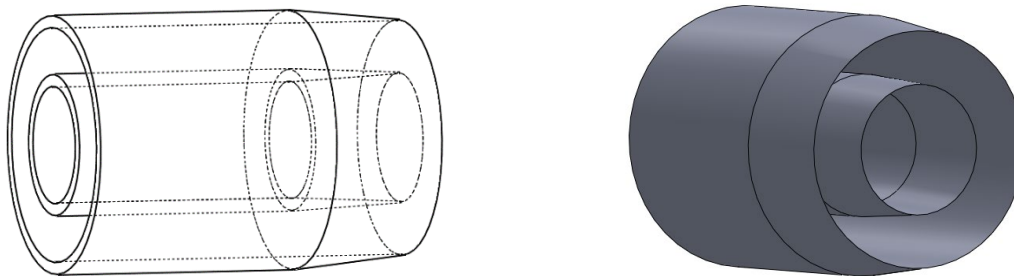


Figure 14. self-build CAD model

3.3. Wedge-shaped deflection method

The methodology used in the wedge-shaped deflection method is running the 3D SST turbulence model first to obtain velocity and turbulence intensity result, then running the DES turbulence model for the accurate SPL result. Because from the previous papers, the RANS model has more agreement with experiment on the potential length, but LES model has more accurate result on testing SPL result. In this part of the research, three models will be simulation, basic model, 40-degree wedge-shape and 60-degree wedge-shape, the number of simulation models is small, more time can be spent on each case, so both SST and DES model will be simulated to obtain the accurate result. Moreover, due to the faster simulation time, the SST turbulence model is used for generating velocity and turbulence intensity contours at the beginning. By analysing characteristics, turbulence intensity and relative velocity between the two jet and potential length of these contours, rough judgement can be make, If changing geometry has no noise reduction effect, no further study is needed, this can reduce a lot of time. In addition, the generated turbulence contour in the SST model can also

be used to locate the noise source's volume and position, then through creating permeable surfaces that surround the source to finish the noise source setting. From the literature review, the permeable should avoid passing through the turbulence as much as possible, the generated turbulence contour can great helpful to the construction of permeable surfaces. Then, a more precise DES model can be applied to accurately measure and generate sound pressure levels.

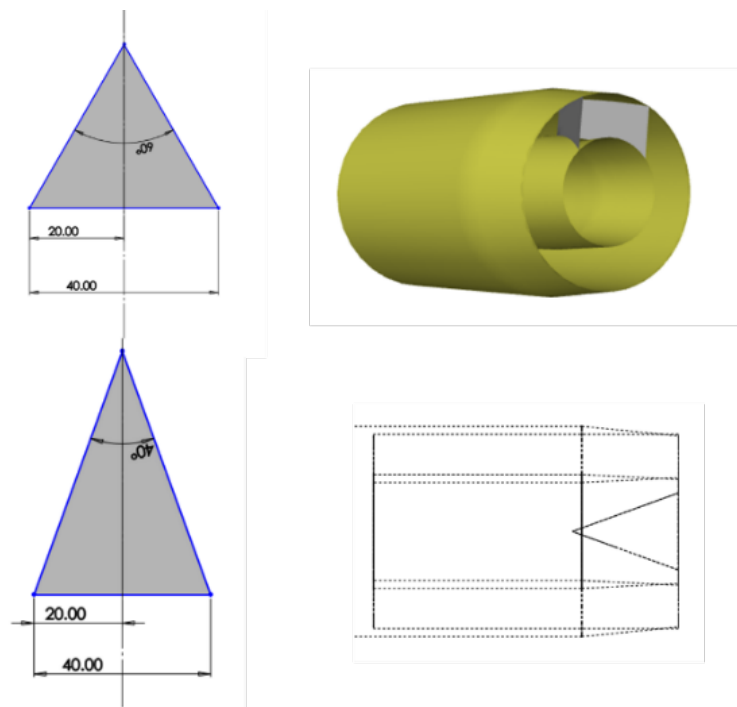


Figure 15. 3D CFD model for fan flow deflection method

The figure shows the 3D CFD model for the fan flow deflection method, the wedge-shaped deflector will insert into the bypass nozzle. The wedge is oriented with its top facing inwards towards the nozzle, while the bottom is in contact with the nozzle lip. Additionally, the upper and lower sections of the wedge come into contact with the walls of the bypass and core nozzle, respectively, and no flow passes through them. Three models will run on the simulations, one model is without wedge as a control group, and the other two are experiment group with wedge, but the angles of wedge are 40 °and 60°respectively and the wedge have the base of the same length of 80mm.

For the mesh, the MultiZone mesh generating method, which can automatic decomposition of geometry into structured regions and unstructured regions, is applied. The simple, structured hexahedral/ prismatic meshes is used in the in the zones 1,2,3,4 in the figure 16 , this can improve the filling volumes efficiency to reduce the element number and increase the convergency ability. But in the zone 5, the unstructured tetrahedral mesh is used, this is because the geometrical structure of bypass and core nozzle is relatively complex and there

is a wedge-shape deflector inserted, the tetrahedral mesh has better adaptability to complex geometrical structure, which improves the accuracy. For the size of the element, small size of mesh is used in zone 5 and 3 to get the better result in the area of jet and ambient air mixture, and the mesh become coarse as distance from the jet-nozzle exit increasing, therefore, in the zone 1, the element size is 0.01m, for the zone 2, the element size is 0.008m, the size of element in zone 3,4 is 0.005m and element size is 0.003m in zone 5.

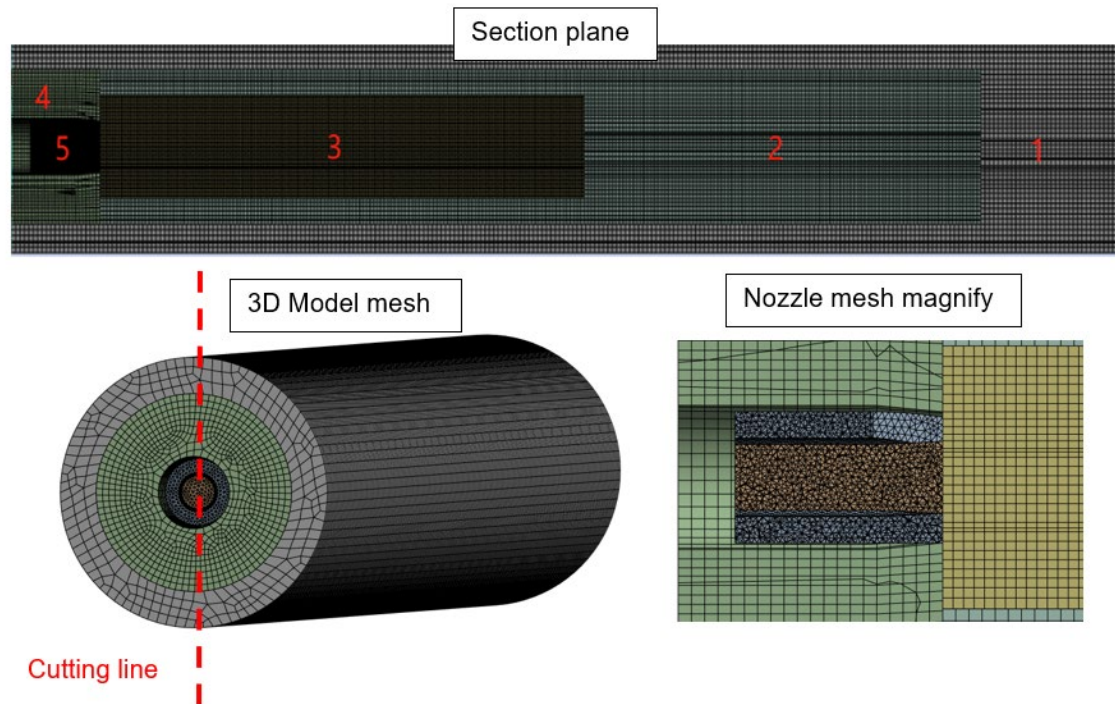


Figure 16. the mesh used in fan flow deflection method

Table 3. the mesh detail for fan flow deflection method

Number	Size (mm)	mesh type
Zone 1	10	hexahedral prismatic
Zone 2	8	
Zone 3	5	
Zone 4	5	
Zone 5	3	tetrahedral

For the boundary condition setting. Considering the take-off speed of aircraft generally is 100m/s, and the bypass and core jet speed are faster than the take-off speed. Therefore, in

the boundary condition setting, the inlets are velocity-inlet type, the velocity of freestream inlet is 100m/s, and velocity of bypass and core flow inlet is 170m/s and 272m/s respectively, The outlet is set at flow exit and the type is pressure outlet. The cold jet is assumed in this case. In addition, the Ffowcs Williams and Hawkings Acoustics Model (FW-H) is applied to measure noise. the close surface method is chosen to encompass the noise source. Depending on the turbulence intensity contour, the region of noise source set as a cylinder with a diameter of 0.2m and a length of 1.4m start at the position of jet exit and noise source is surrounded by permeable surfaces to reducing the waste of computation resource. The receivers are installed on an arc with the jet outlet as the center radius $r = 100D$, the detail position shows in table 5.

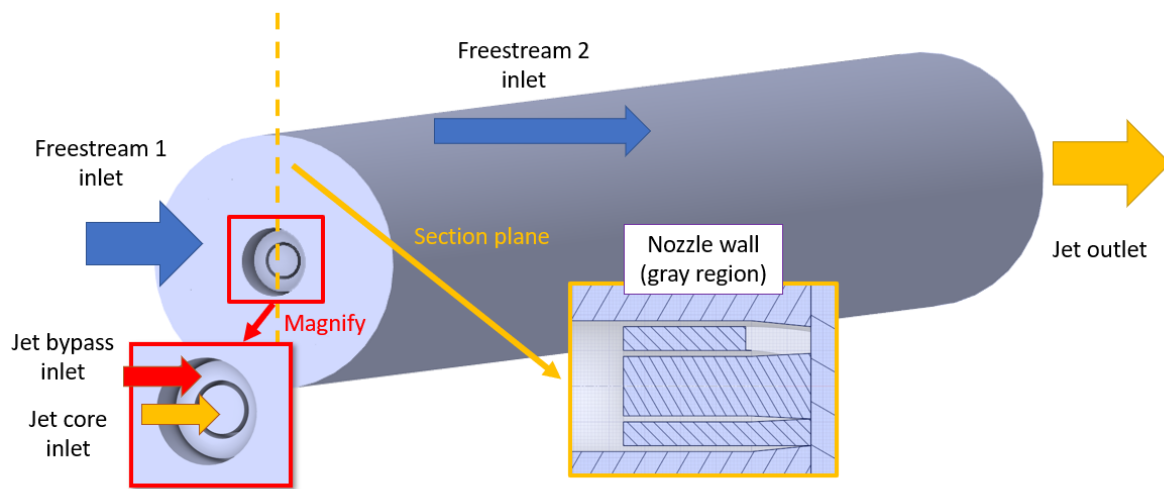


Figure 17. the boundary condition in fan flow deflection method

Table 4. Boundary condition set in fan flow method

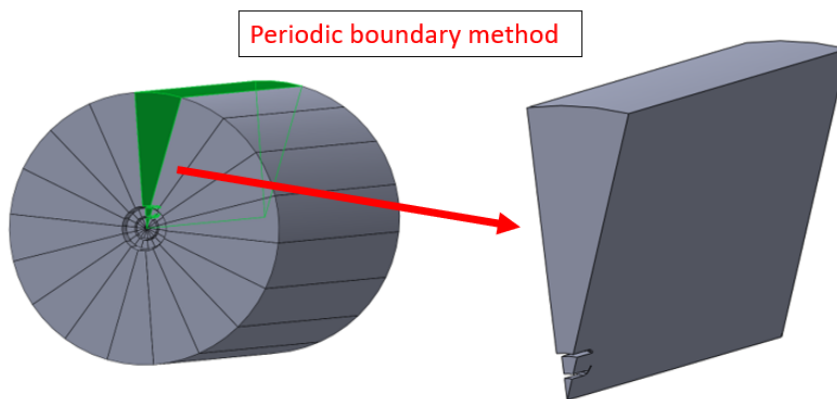
Name of Boundary Condition	Type	Value
Freestream 1 inlet	Velocity inlet	100 m/s
Freestream 2 inlet	Velocity inlet	100 m/s
Jet bypass inlet	Velocity inlet	170 m/s
Jet core inlet	Velocity inlet	272 m/s
Jet outlet	Pressure outlet	101325 Pa
Nozzle Wall	Wall	NA

Table 5. receivers' position (the center of jet exit as origin position)

Direction angle	X-position (m)	Y-position (m)
30 degree	8.57	4.95
60 degree	4.95	8.57
90 degree	0.00	9.90

3.4. Chevrons method

For this part of the research, the effect of chevrons nozzle design on noise reduce will be explored. Specifically, the emphasis will be placed on the application of chevrons on the core nozzle, and the two affect factors, chevron length and the penetrate angle will be simulated by the Ansys. For the chevron length, assuming the bypass chevrons length will be 20mm, then varying core chevron lengths to simulate the result, analyzing the velocity, turbulence and acoustic performance of the chevron nozzle design in terms of reducing noise levels. For the penetrate angle. The 20-degree and -20-degree of penetrate angle will be stimulated, this penetrate angle is very large and is impractically for the real world. However, it can enhance the noise reduction level and a clearly result in noise can be observed to analyse reduction effect by turning the core chevrons up and down. The turbulence model chosen strategy is similar to strategy used in the fan flow deflection, using the SST turbulence model first to obtain the initial result, then DES turbulence model will be utilized to further investigate. Considering both chevron length and penetrate models are symmetric models, so the model is divided into 18 equal parts, and using the periodic method, which defines a cyclic situation of the flow across the boundary surface, to simulate the models, this can save a lot of computation resource and reduce the simulation time.

**Figure 18.** the periodic method used in the model

For the CAD models used in the simulation, the following variables in the length and penetrate angle models is same, the number of the chevrons is 18 for both core and bypass nozzle and the position of the chevrons in the core and the bypass nozzle is staggered and the chevron length of the bypass nozzle is 20 mm. the 4 chevron length models is built with the different

core nozzle chevrons length, 0mm, 20mm, 30mm and 40mm. For the penetrate angle case, two models are built, by altering the orientation of the chevrons at 20 degree and -20 degree, and the chevron length of the core nozzle will be 20 mm.

Table 6. the detail information about the models

Model	Chevron length bypass nozzle	Chevron length core nozzle	Penetrate angle
Model 1	20 mm	0 mm	0 °
Model 2	20 mm	20 mm	0 °
Model 3	20 mm	30 mm	0 °
Model 4	20 mm	40 mm	0 °
Model 5	20 mm	20 mm	20 °
Model 6	20 mm	20 mm	-20 °

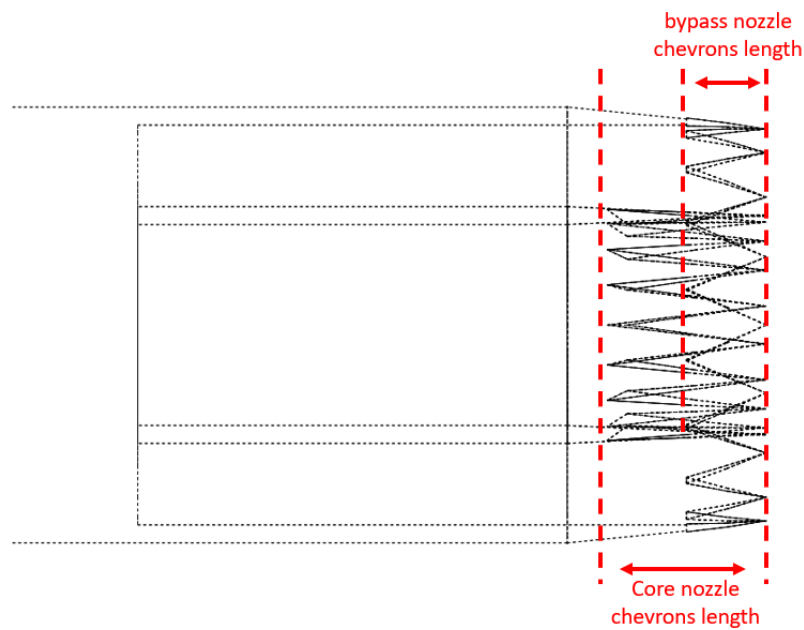


Figure 18. the position of the core and bypass chevrons length

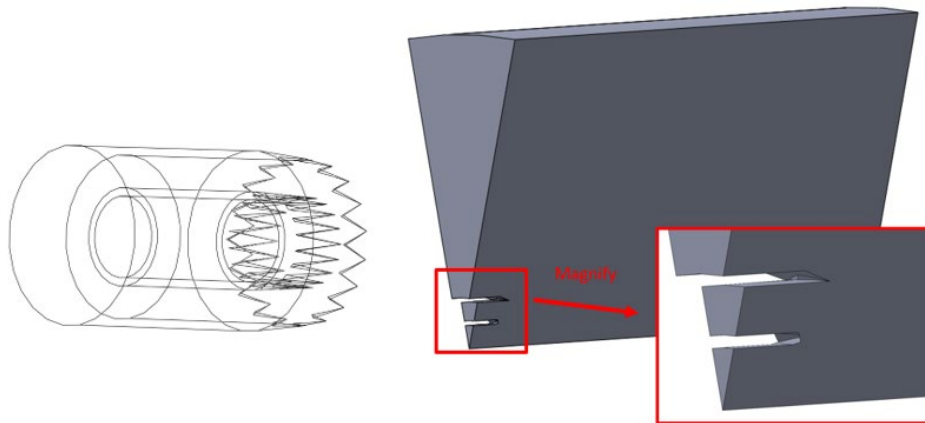


Figure 19. the chevron length model

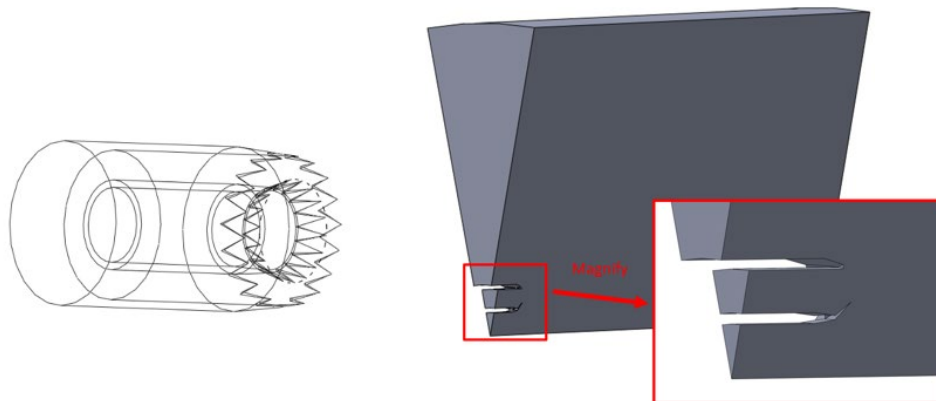


Figure 20. the chevron penetrate angle at 20 degree

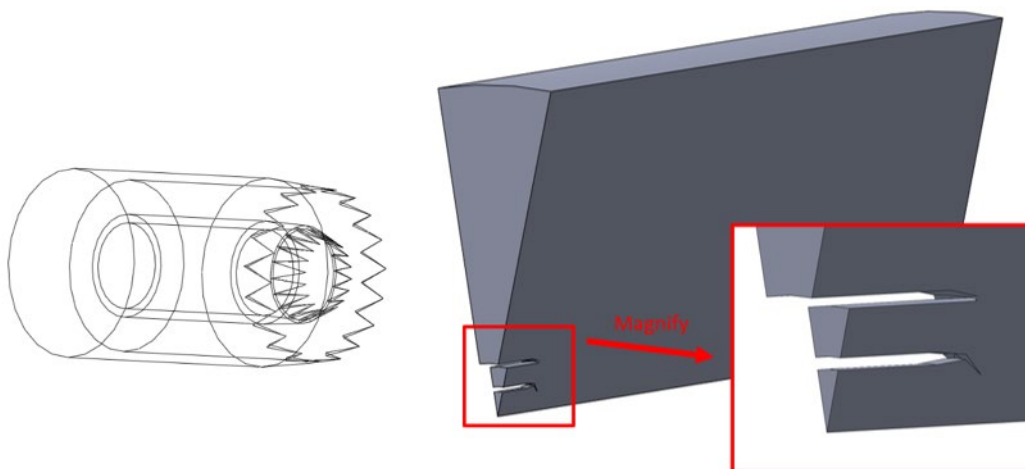


Figure 21. the chevron penetrate angle at -20 degree

For the mesh, the strategy used in the chevron method is quite similarly as the mesh strategy used in the fan flow deflection method and same strategy is used for both chevron length and the penetrated angle cases. The MultiZone mesh generating method is used. For the core and bypass nozzle, the geometry is complicated and the mesh should show the details of the chevrons geometry, so the unstructured tetrahedral mesh with the 3 mm element size is used. The zone 4 in the figure 23, where the jet mixed with ambient flow also requires high quality of mesh, therefore, the 5 mm structured hexahedral/ prismatic meshes are used, for the other region in the figure 23, the region geometries are regular and have minimum effect on the simulation result, the meshes of these regions are structured hexahedral/ prismatic meshes, and the size in zone 2,3 and 5 are 8 mm, 8 mm and 10 mm. Additionally, to keep both sides of the mesh the same for the periodic boundary requirement, the match control method is used.

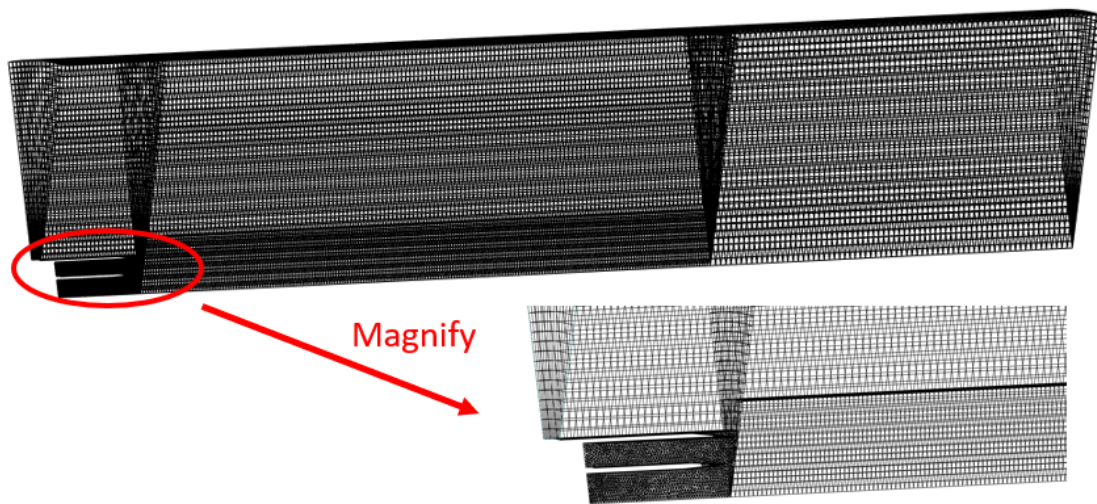


Figure 22. the mesh used in chevrons method

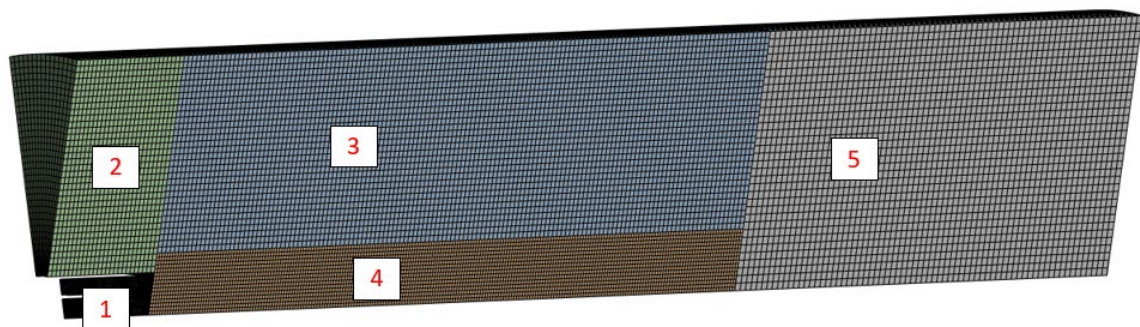


Figure 23. the different mesh region position in chevrons method

Table 7. the mesh detail for chevrons method

Number	Size (mm)	mesh type
Zone 2	8	hexahedral prismatic
Zone 3	8	
Zone 4	5	
Zone 5	10	
Zone 1	3	tetrahedral

For the periodic boundary setting, after the using the match control method, the Fluent will automatically setting the periodic boundary, so during the simulation, the flow can across this boundary surface. The table 8 shows the detail about the boundary information in the models, the inlet is velocity inlet type and the outlet is pressure outlet type. The permeable surfaces is surround the zone 4 in the figure 23 as the noise source. This would be used after completing the DES turbulence calculation and using FH-W approaching to obtain the acoustic result. To obtain the far-field noise, the distance of the receivers are 100 nozzle diameters from jet exit position and the receiver positions are the same as the fan flow deflection method and shows in table 5.

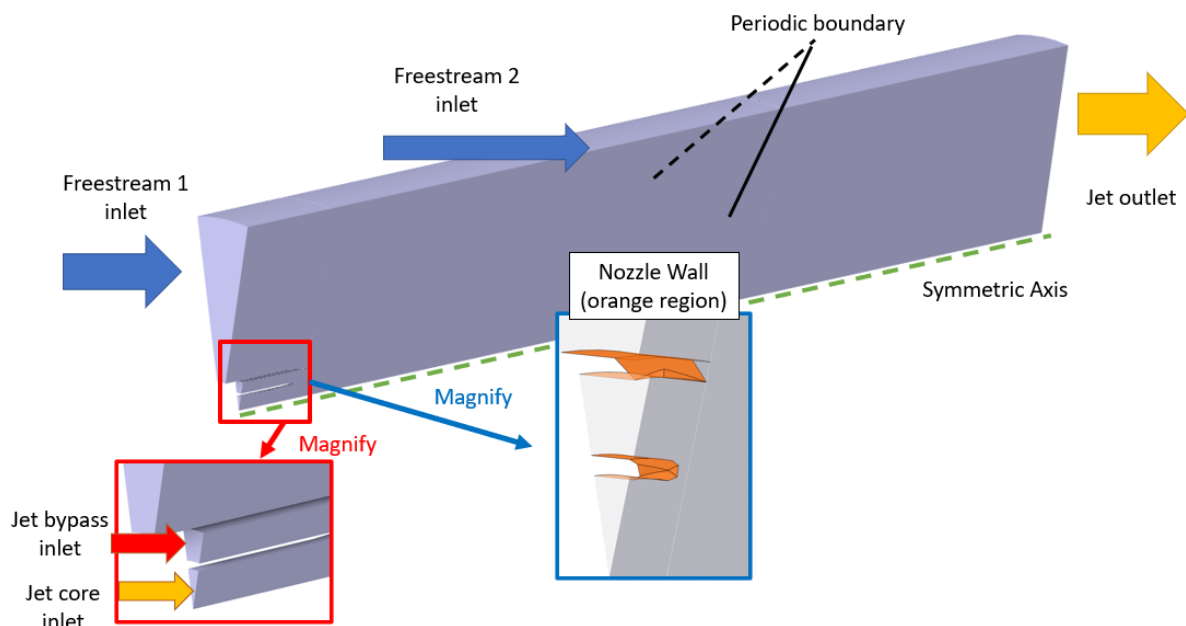
**Figure 24.** the boundary condition in chevrons method

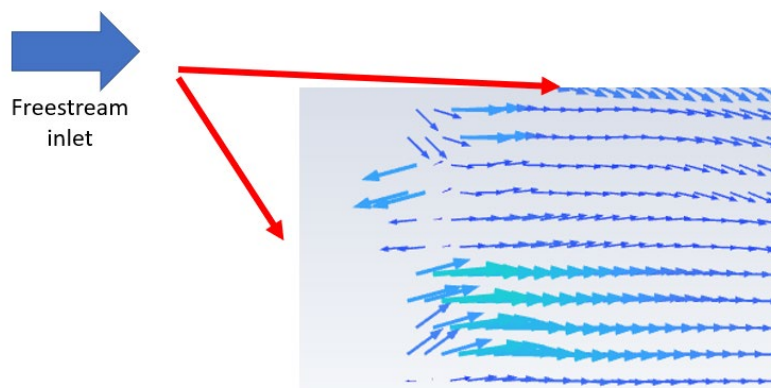
Table 8. Boundary condition set in chevrons method

Name of Boundary Condition	Type	Value
Freestream 1 inlet	Velocity inlet	100 m/s
Freestream 2 inlet	Velocity inlet	100 m/s
Jet bypass inlet	Velocity inlet	170 m/s
Jet core inlet	Velocity inlet	272 m/s
Jet outlet	Pressure outlet	101325 Pa
Nozzle Wall	Wall	NA
Periodic boundary	Periodic	NA
Axis	Axis	NA

4. RESULTS AND DISCUSSION

4.1. CFD Validation

For the mesh, it is clearly that coarse mesh can't not get the correct result, but if the mesh size is too small, the result of the continuity on the residual plot can not convergent and . In addition, on the transient case, it will cause time step very small, 10^{-6} in my case. And compare with the time step 10^{-4} , the speed is much slower. Therefore, the number of elements is changed from 355200 to 106102 and the simulation result remain unchanged. For the inlet freestream boundary condition. the left and top boundaries can not select together as one velocity inlet, otherwise the freestream will reverse near the boundary wall, the detail velocity vector shows below. And the set the two boundaries separate can solve this problem.

**Figure 25.** Error vector contour about freestream reverses near the boundary

For the plot of velocity changed along the radial axis at the position of $x/D_j = 2$ and 5 with the velocity of 0.51 Ma. Comparing steady SST, transient SST and DES model with the NASA experiment result, the CFD result is nearly same as the experiment results. all turbulence models can be used. Comparing with the $x/D_j = 2$ and 5 plot, the turbulence velocity reduces much quicker on the radial direction as the axial position is closer to the nozzle.

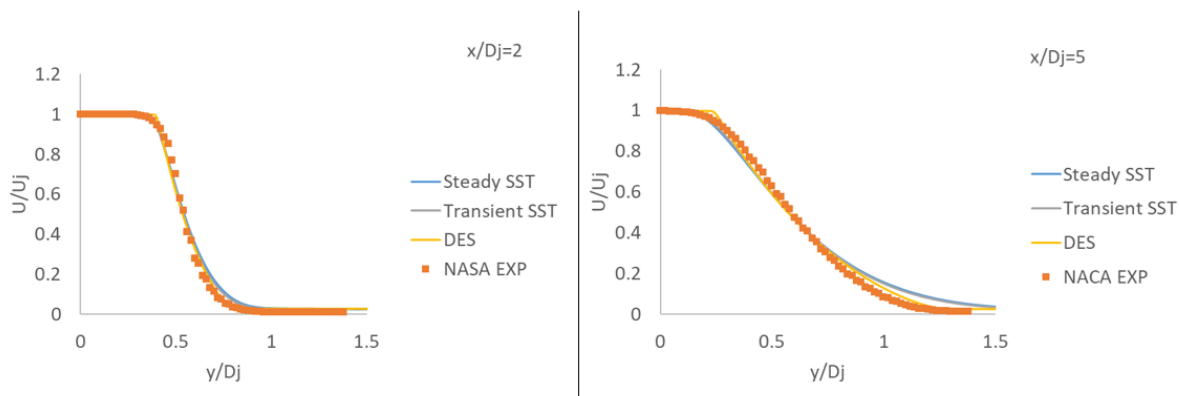


Figure 26. Axial turbulent velocity in radial profiles at $x/D_j = 2, 5$ position

On the plot u/U_j VS X/D_j along the centreline at the condition of Ma is 0.51 and 0.99. It can be found from the plots that as the jet exit velocity increases, the position of the velocity separation point becomes farther away. The separated position of $x/D_j = 2$ is at 6 and the $x/D_j = 5$ is at 8. For the DES turbulence model, the velocity changing point position is almost same as the position in the experiment, and with velocity increased, the changing point position is far away from the nozzle where jet exit. However, for both steady and transient SST turbulence model, the velocity changing position will be behind the experiment results in the velocity of 0.51 and 0.99 Ma. Although the SST turbulence model is a little behind than the experimental separation point, the separation point change distance between the two speeds is almost the same as the experimental one, which is 1 X/D_j . So all the turbulence models meet the requirement.

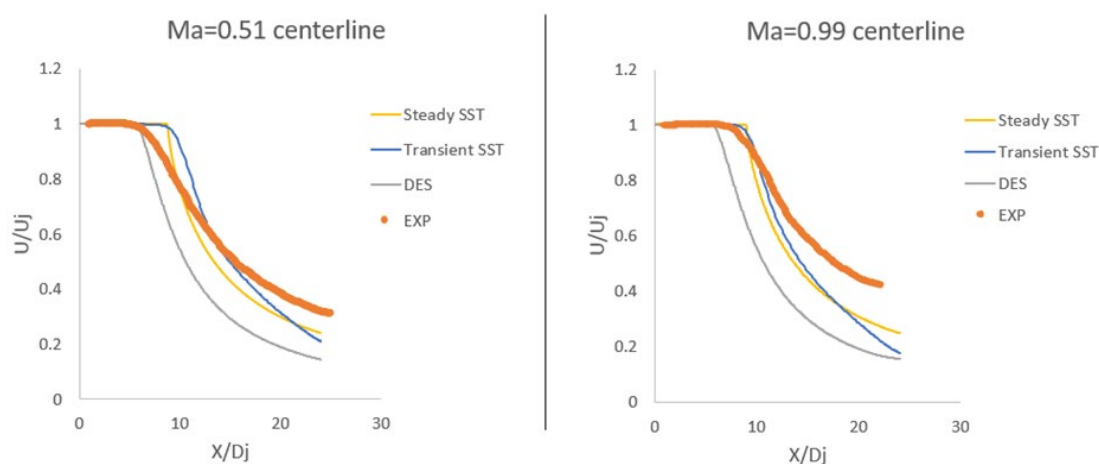


Figure 27. Axial turbulent velocity in axial profile along centerline with different turbulence model

For the turbulence profile at distance 8 times of nozzle diameter away from nozzle exit at the centerline with the velocity of 0.51 and 0.99 Ma, the steady SST model was analysed. The overall trend of CFD result is meeting with experiment result. the difference is at the beginning of CFD result, the curve is around 0 until reaching the certain distance behind the nozzle, the curve increases shapely. But for the experiment result, the curve rises slowly at the beginning and then increases dramatically. In addition, for the peak point position, the k/U_j^2 value is same for both velocities, but the location for higher speed 0.99 Ma reaches the peak point is close to the nozzle and the CFD result peak point value is bigger than the experiment result. In general, the steady SST model can be used. For unsteady case, because ANSYS Fluent doesn't have mean turbulent kinetic energy option, as a result, plot of turbulent kinetic energy oscillates up and down. Therefore, the plot is not given in unsteady case. But considered turbulent kinetic energy is calculated by the equation $k = \frac{1}{2}(\overline{u'^2} + \overline{v'^2} + \overline{w'^2})$. The turbulent kinetic energy is related to the velocities of 3 directions and according to velocity validation and transient case is more accurate than the steady case, assumed that unstable situations can be used.

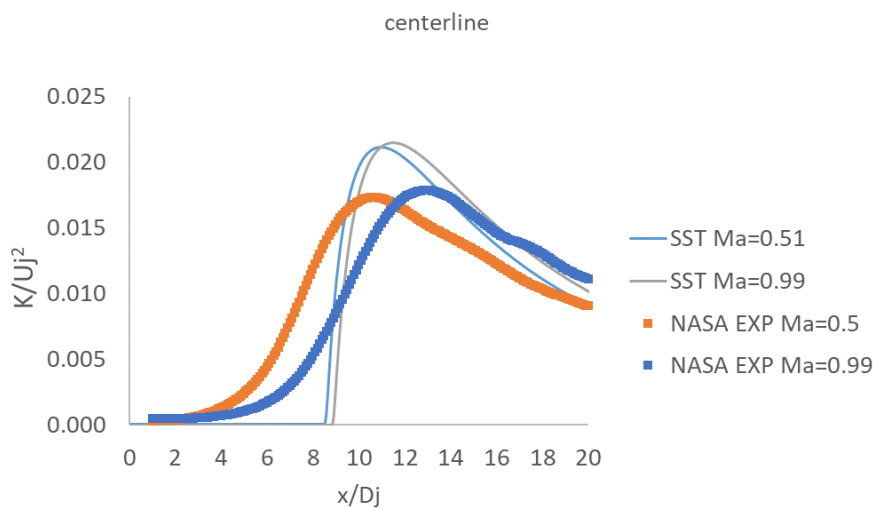


Figure 28. k/U_j^2 in axial profile along centerline

4.2. Wedge-shaped deflection method

The provided figure illustrates the flow performance on the wedge wall and core nozzle wall. It indicates that the flow near the wedge initially moves parallel to the other flows but subsequently tilts downwards, this means the wedge-shaped deflector has a turning effect.

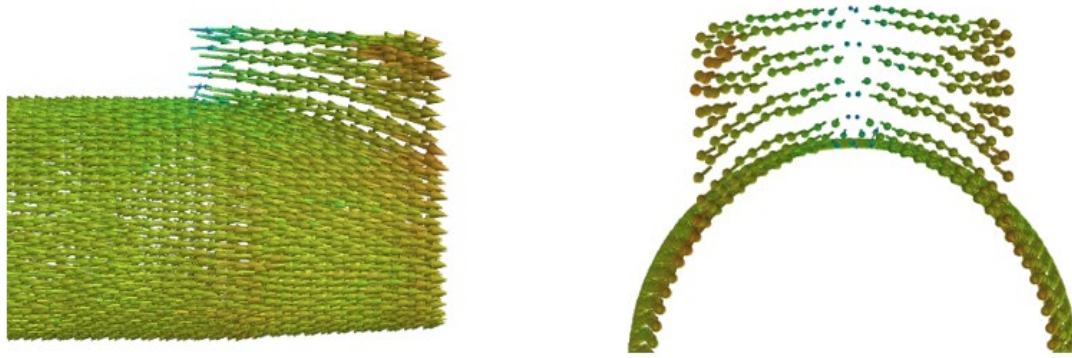


Figure 29. Velocity vector on wedge wall

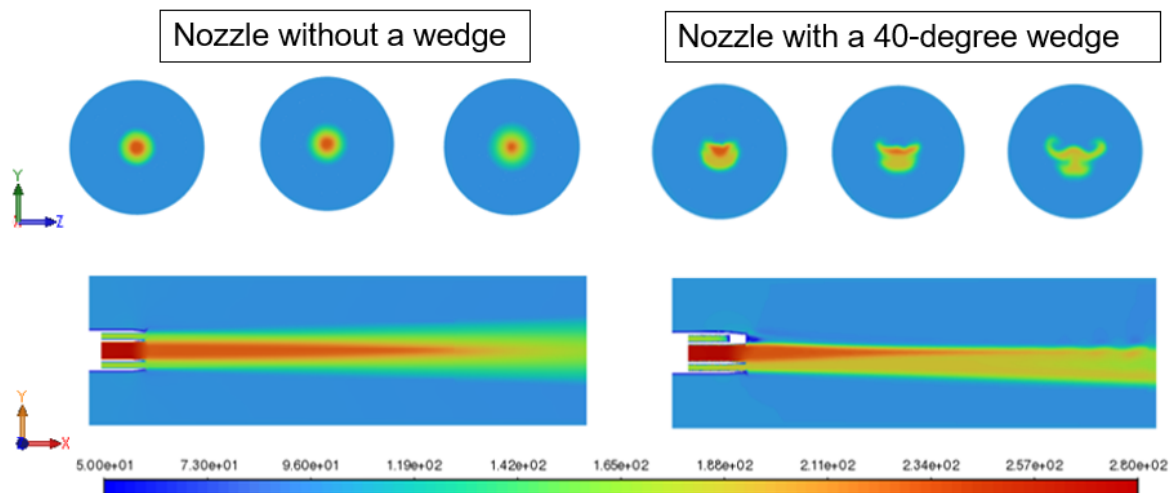


Figure 30. The contours of velocity at $x/D=2,4,8$ and xy cross section 40 degree

The figure illustrates velocity contour with two types of nozzles. It is worth noting that the general trend of velocity contour in the 60-degree wedge is comparable to that in the 40-degree wedge, so it is meaningless to compare with them and the figure doesn't give. But the detail bypass jet velocity change will be showed after this. According to the velocity contour at the xy cross section, the nozzle without a wedge shows there is no discernible difference between the bypass jet on the upper and bottom sides of the core jet. However, for the nozzle with the wedge, the velocity contour shows a significant difference between the bypass jet on the upper and bottom sides of the core jet. the bypass jet on the upper side of the core jet is notably weaker, while the bypass jet on the lower side is wider and stronger. This difference is coming from the interaction between the wedge and the jet. The wedge turns upper bypass jet downside and produces an asymmetric distribution of velocity in the bypass jet and a thick low-speed layer was generated, which is satisfied fan flow deflection (FFD) noise reduce theory. According to the velocity contour at the yz cross section, the velocity contours is similar

at the $x/D=2,4,8$ for nozzle without wedge. The different is the core jet region become small and outer ring is occupied by the bypass jet region with the distance increases from the nozzle. For the nozzle with wedge inserted. The shape of the velocity contour is like bullhead shape, the velocity section has a trend to extend sideward and upward. In addition, for both nozzle with the distance from the nozzle increases, the boundary of the velocity field gradually becomes indistinct.

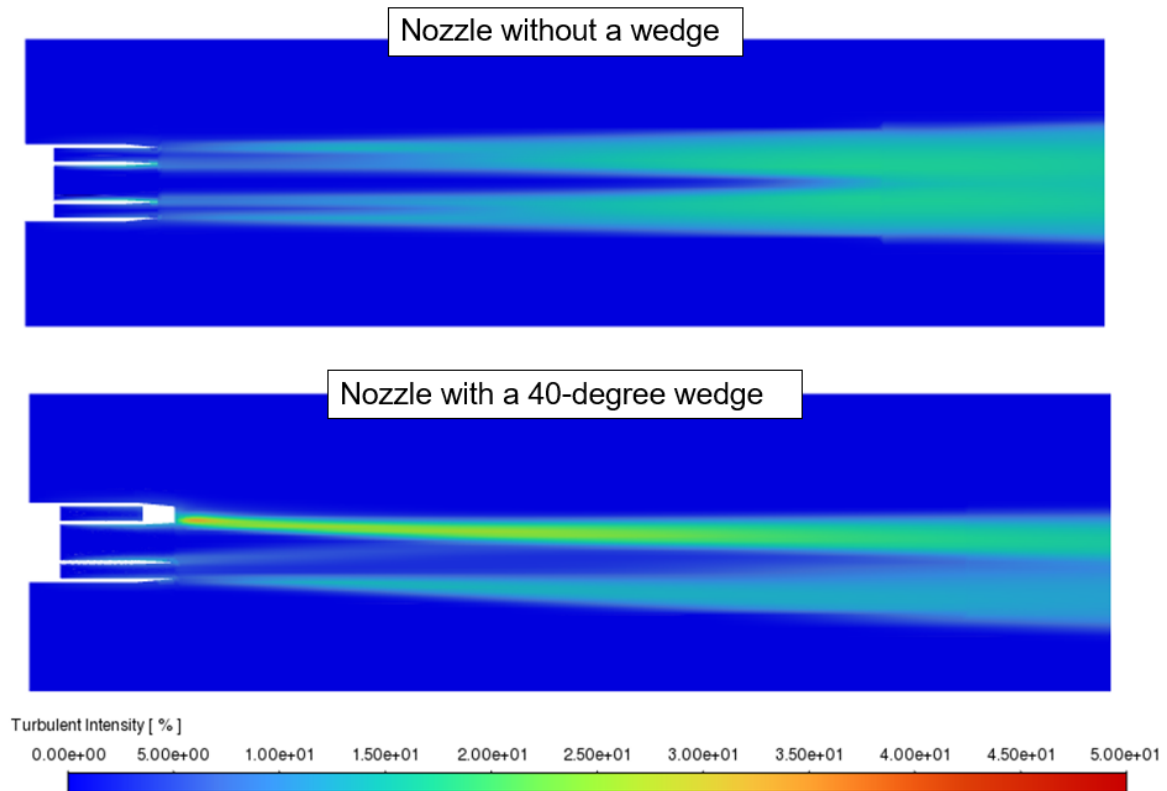


Figure 31. The contours of turbulent intensity

The figure 31 above indicates the turbulent intensity contour after bypass and core jets exit the nozzle. The nozzle without wedge shows that there is a long and wide plume behind core nozzle and two short and thin plumes behind bypass nozzle and the turbulent intensity in these plumes is low and high turbulent intensity layers surround these low-intensity plumes. Additionally, the turbulent intensity increases as the distance between the nozzle increases. For the nozzle with wedge, the largest turbulent intensity is behind the wedge and the high frequency noise may increase at this region. Because of the wedge, the upper low turbulent intensity layer almost disappeared. The plume behind the core nozzle still exists, but the length of the plume is much short comparing with the nozzle without wedge and the bottom high turbulence intensity layer is weak. For the region behind the bottom bypass nozzle, the low turbulent intensity area becomes wider at first and then at the point where the plume behind the core nozzle disappeared, the area starts to narrow. In general, the high turbulent intensity layers around the plumes are the main noise source. comparing these layers with nozzle with and without wedge, the turbulent intensity layer between the upper bypass jet and ambient flow disappeared in the wedge nozzle model. although higher turbulent intensity layer occurs

at upper region of core jet, but its turbulence region is much wider. For the nozzle with wedge, the turbulent layer between the core and bottom bypass jet is weak.

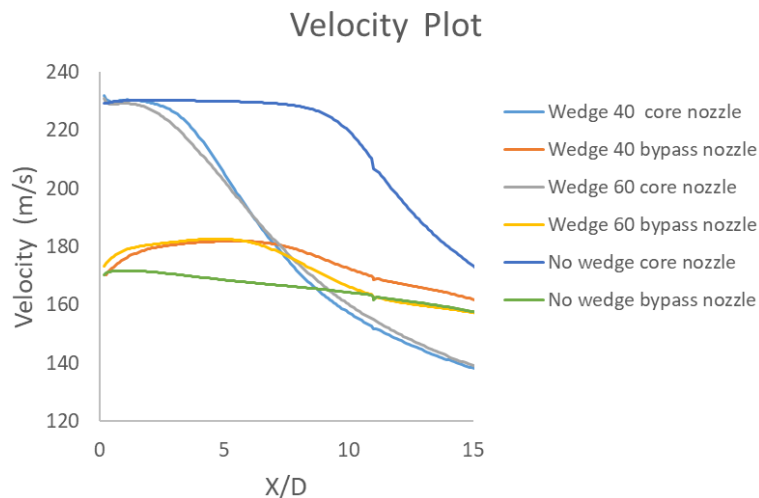


Figure 32. The core and bypass velocity plot

For the velocity plot, the main focus is to compare the velocity difference at the jet exit of the core nozzle and the bypass nozzle. As mentioned in the literature review, the large-scale eddies are generated at the shear layer of turbulence and treated as noise source to emits instability waves. the level of this noise is related to relative speed of two jets, called as convective Mach number. Reduce the convective Mach number can weak the emission of the instability waves to reduce noise. For the core jet exit from the nozzle without wedge, for a long time, the velocity remains constant until $X/D=8$. The bypass jet has been slowly decreasing after leaving the exit. For the wedge nozzle, due to the upper flow turns downwards, the core jet velocity reduces immediately after exit from nozzle at the position about $X/D=2$. Due to the addition of upper bypass jet, the bottom bypass jet increase at first from 170 m/s to 180 m/s then velocity decreases. Therefore, the nozzle inserting wedge has smaller area between the core and bypass jet velocities than that nozzle without wedge. It is worth noting that the detected velocities are from two parallel lines passing through the center of the core and the bypass nozzle, the second half of the detected core jet has been mixed with the ambient flow due to the turning effect, so bypass jet is faster than core jet. Comparing with 40-degree wedge and 60-degree wedge, the 60-degree wedge has great turning effect. Its core jet changes velocity quicker than the 40-degree wedge and its bypass jet reduce much faster than the 40-degree wedge.

From figure, the general trend for the acoustic field spectra at 30-degree and 90-degree is high sound pressure level at the low frequency and low sound pressure level at high frequency, and the sound pressure almost keep constant at the high frequency. For the sound spectra at 30-degree. The frequency range between 200 Hz and 800 Hz significantly contributes to the perceived noise level. From this range, the peak node of the no wedge

nozzle is at 71dB, and the 40-degree and 60-degree wedge reduce about 10 and 13 dB respectively. The sound pressure level for high frequencies is nearly identical for the nozzle with and without a wedge. For the sound spectra at 90-degree, the peak value of the no wedge nozzle is 56 dB and the peak value of 40-degree and 60-degree is about 48dB. 8dB has been reduced because of the wedge. In addition, for the sound pressure level in high frequency is main at about 24dB. Comparing with spectra at 30 and 90 degree. the sound pressure at the 30-degree is higher than that at 90-degree at the low frequency and almost same at the high frequency. For the noise reduce effect of wedge-shape deflector, more noise reduces at the 30-degree spectra and the large angle wedge reduce more noise. In addition, in the CFD post section, the fast fourier transform is used to transform time into frequency and Hanning type signal filter is used to filter noise.

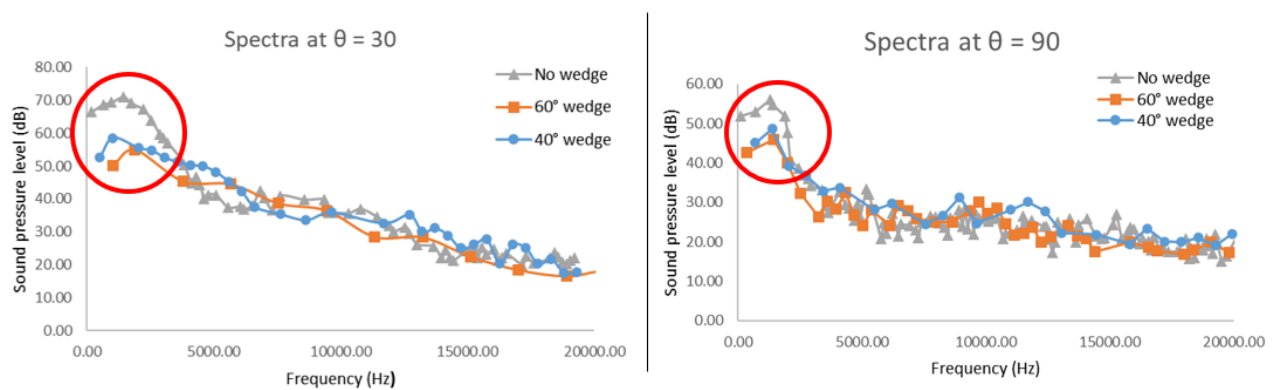


Figure 33. The sound pressure level at 30° and 90° degree direction

The table shows the overall sound pressure levels measured at three different directions 30°, 60°, and 90° for three different nozzle configurations: no wedge, 40° wedge, and 60° wedge. According to the results, it can be inferred that the wedge-shaped deflector with a larger angle significantly reduces noise at a measurement direction of 30°. However, as the measurement direction shifts downwards, the extent of noise reduction decreases.

Table 9. Overall sound pressure level

Direction	No wedge	40° wedge (dB)	60° wedge (dB)
30°	121.58	117.78	116.19
60°	117.54	114.01	114.73
90°	115.89	112.37	110.64

4.3. Chevrons method

4.3.1. Varying chevron length in core nozzle

For the flow field, the figure 34 shows that axial velocity plot at the centerline with different core nozzle chevron's lengths. In the figure, the velocities at the bypass nozzle almost the same, this shows that the changing the chevrons lengths in the core nozzle doesn't influence the velocity at bypass nozzle, if the chevron's length in the bypass nozzle doesn't change. However, the velocity lines in the core nozzles have some different. Although the U/U_j is around 1 at nozzle exit, the jet exit velocity increases as the chevron's length increases and the U/U_j changes from 0.97 to 1.04. Because of the vibration, the potential length of core nozzle is hard to identify, the centerline velocity decay shows that the long length of chevrons reduce velocity faster, but at the position of $X/D_s=15$, the velocities with varies lengths are same at $U/U_j=0.67$.

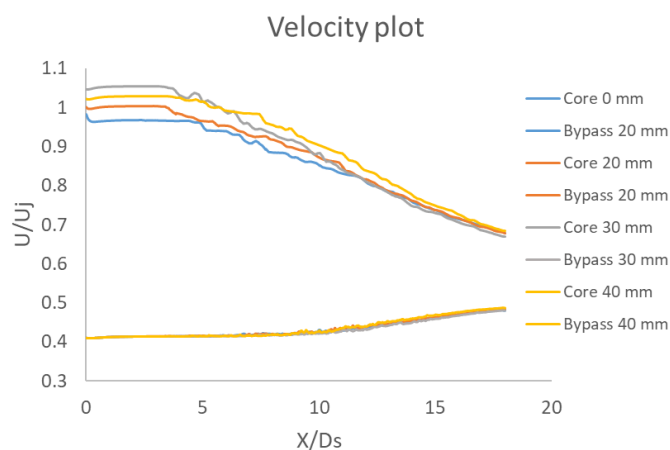


Figure 34. The axial velocity plot for varying chevrons length at centerline

For the acoustic field, the figure 35 indicates the SPL spectra at the emission angle of 30° and 90° . The SPL spectra gives the information that changing the length can reduce the low-frequency noise with the consequence of increasing the high-frequency noise. At the 30° emission angle, the low-frequency noise level is around 50 dB and the high-frequency noise is nearly at 15 dB. At the 90° emission angle, the low-frequency noise level is about 40 dB and the high-frequency noise is approximately 15 dB. It seems that the low emission angle has high noise at low-frequency, but there is no difference between the low and high emission angle at high -frequency. In the table 10, the overall sound pressure level is listed, the 0 mm chevron length at 30° has loudest noise, and the 40 mm chevron length at 90° has lowest noise, this also proves that more noise is emit at the low emission angle and the long chevron length can reduce more noise. The result is actually different with the Bridges and Brown [18] experiment result. In their experiment, the length of chevrons has little impact on the sound. The types of nozzle is one of the possible reason, in the simulation, the nozzle used is dual jet nozzle but in their experiment is single jet nozzle. Furthermore, in their experiment excepting, changing the length of chevrons, some other parameters is slightly modified to keep

the vortex strength parameter. In addition, the Fluent constraints the number of detecting data, and small time steps applied in the simulation, both of these cause the number of data in the low-frequency is not enough.

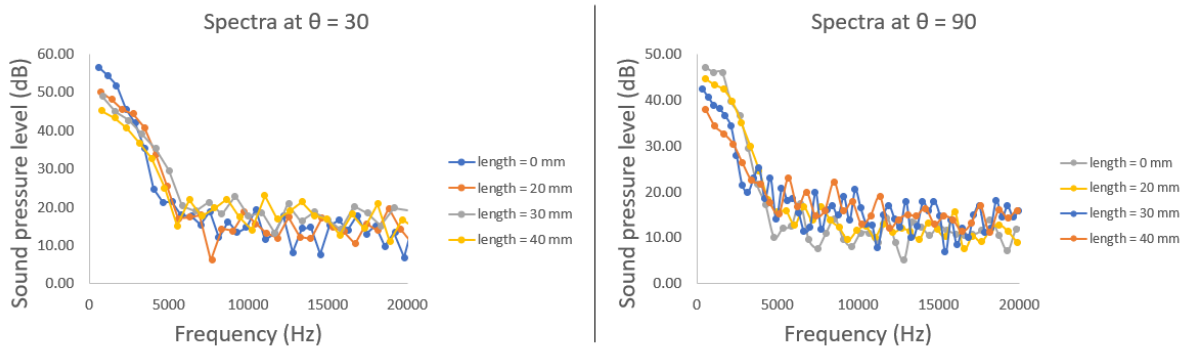


Figure 35. The sound pressure level of vary length at 30° and 90° degree direction

Table 10. Overall sound pressure level

Direction	Chevron length 0 mm (dB)	Chevron length 20 mm (dB)	Chevron length 30 mm (dB)	Chevron length 40 mm (dB)
30°	116.49	115.87	114.29	113.60
60°	115.59	114.22	112.84	110.93
90°	113.78	112.23	111.53	110.84

4.3.2. upwards and downwards chevron penetration in core nozzle

For the flow field, the axial velocity plot in the figure shows that the inwards chevrons angle can increase both core nozzle and bypass jet, the U/U_j values are 1.3 and 0.6 and outwards chevrons can reduce the core nozzle jet, the U/U_j is 0.8 and increase the bypass nozzle jet, the U/U_j is 0.6. The potential lengths for both core and bypass nozzle at penetrated angle at 20° are decreased and the velocities reduce rates is much faster than that penetrated angle at 0°, this can reduce the noise source length and reduce the noise. but the velocity difference between the core and bypass nozzles has increased. For the velocity lines at the -20° penetrated angle, the potential lengths are increased, this is not good for the noise reduction, but velocity difference between two nozzles is reduced, and this may reduce the turbulence generation and reduce the noise.

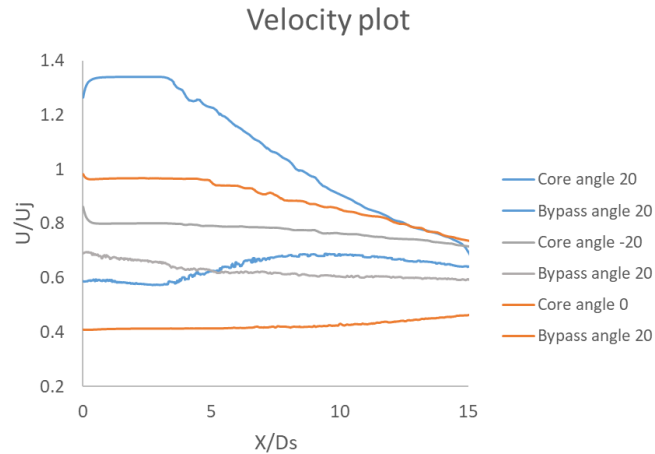


Figure 36. The axial velocity plot for penetrated angle

For the acoustic field, the spectra at 30° , both -20° and 20° angle cases have reduced the low-frequency noise. The emission angle at 90° , the 20° penetrated angle reduced the low-frequency noise, but the -20° penetrated angle, the low-frequency noise doesn't increase or decrease. In addition, for the high-frequency noise, the penetrate angle of chevrons has little effect on this. The overall SPL in table 11 shows that 20° angle has some effect on noise reduce, it only reduces about 1dB noise. However, -20° angle chevrons has no noise reduce effect. The large chevron angles may be the reason to cause this, Bridges and Brown [18] experiment shows that the small penetrated angle has good noise reduce effect, some small penetrated angle should be tried on the simulation.

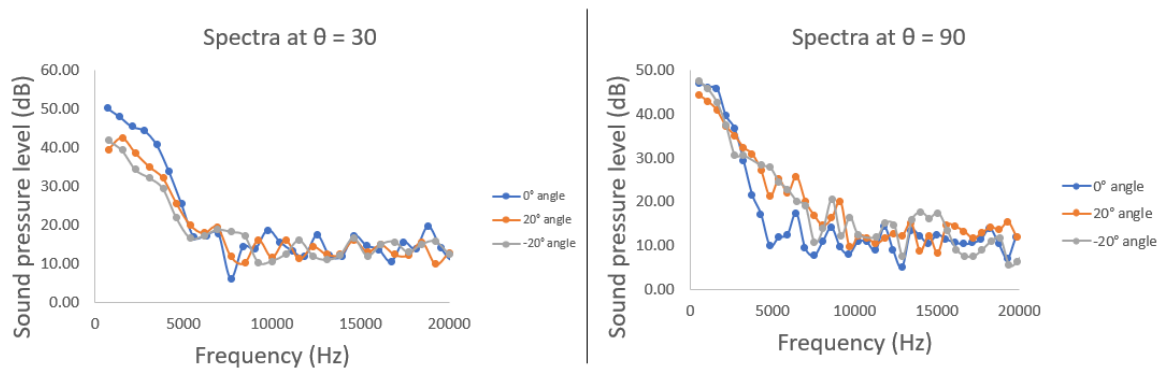


Figure 37. The sound pressure level of vary length at 30° and 90° degree direction

Table 11. Overall sound pressure level

Direction	Chevron angle 0° (dB)	Chevron angle 20° (dB)	Chevron angle -20° (dB)
30°	115.87	113.87	115.91
60°	114.22	114.07	114.85
90°	112.23	111.26	112.16

5. CONCLUSIONS

To summarize, this project is depended on the CFD method and can be divided by the three main parts, CFD validation, wedge-shape deflector nozzle and chevrons. For the first part CFD validation, the different strategies in the simulation including, 2D or 3D geometries, mesh quality, turbulence models and boundary conditions are explored, in the second part, the 40° and 60° wedge-shaped deflectors are simulated in the Fluent, and results show that wedge deflector can tilt the bypass jet downwards to reduce noise and 60° wedge-shaped nozzle has good noise suppression. In the final part, the chevron's length and penetrated angle in the core nozzle have been studies, the result shows that length of chevrons can influent noise reduction and long chevrons has great suppression effect. For the angle of chevrons, 20° angle of chevrons has sightly noise reduce effect but the 20° angle has no contribution. During the project, the following factors may influence the result. Few low-frequency noise data is detected on the receivers and the close-surface is used to surround the noise source, but this can let the turbulence pass through the surface. Additionally, the simulation is based on the cold jet, but in the real case, the hot jet should be used.

6. FUTURE WORK

Regarding future work, there are still some issues that need to be addressed in the project. Specifically, simulating small penetration angles will be necessary to investigate their noise suppression effect, as the angles used in the project were too large. Additionally, it may be necessary to replace the cold jet with a hot jet to obtain more accurate simulation results. Furthermore, research can be conducted on the installation noise caused by the jet engine and wing.

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8. APPENDIX 1 – Other relevant material

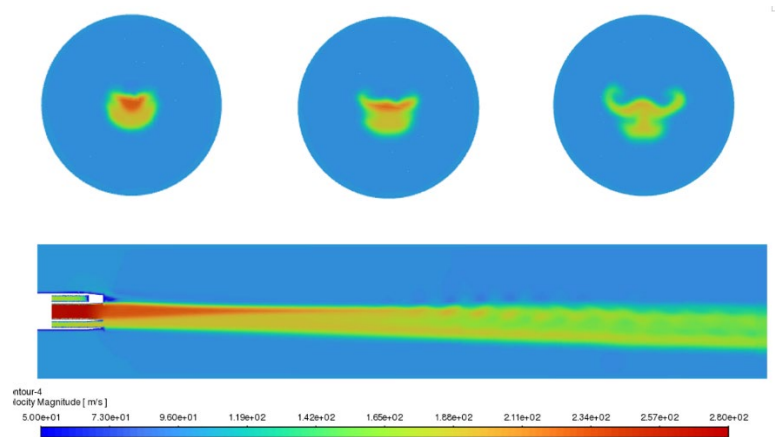


Figure 38. Velocity contour at $x/D=2,4,8$ and xy cross section 60 wedge nozzle

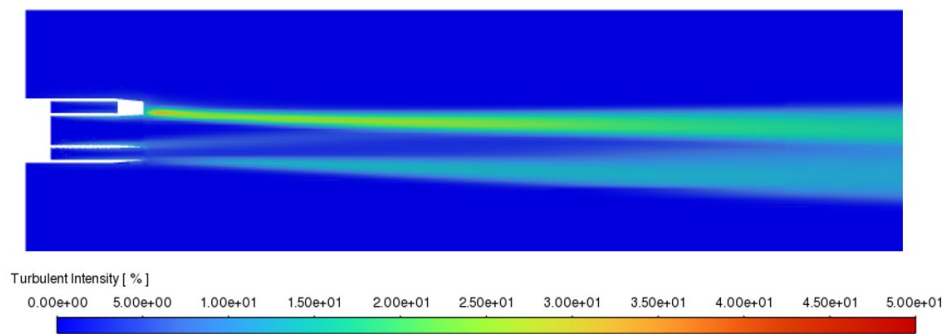


Figure 39. The contours of turbulent intensity for 60 wedge nozzle